

BobaLink Browser Extension

Installation Guide

Version: 1.0.0

Date: 2026-06-06

Classification: Internal

Table of Contents

1. Prerequisites
2. Browser-Specific Installation
3. Manual Installation
4. Post-Installation Setup
5. Configuration Guide
6. Verification
7. Updating
8. Uninstallation
9. Troubleshooting Installation Issues

1. Prerequisites

1.1 Infrastructure Requirements

Before installing BobaLink, ensure the following backend infrastructure is in place:

Option A: Boba Project (Recommended)

The Boba orchestration server provides the full feature set including search aggregation, streaming, and auto-discovery.

Component	Requirement	Verification Command
Docker Engine	24.0+	<code>docker --version</code>
Docker Compose	2.20+	<code>docker compose version</code>
Available ports	8443 (Boba HTTPS)	<code>netstat -tlnp grep</code>
SSL certificate	Valid or self-signed	<code>curl -k https://localhost</code>

Quick verification that Boba is running:

```
# Using curl (replace with your Boba server IP)
curl -k https://192.168.1.100:8443/api/v1/health

# Expected response:
# {"status": "healthy", "version": "1.4.2"}
```

Component	Requirement	Verification
qBitTorrent	4.4.0 or later	Check in qBitTo
WebUI enabled	Configured and running	Access <code>https</code>
HTTPS access	TLS certificate (self-signed OK)	Browser can rea
Authentication	Username/password configured	Login page acce

Enabling qBitTorrent WebUI:

- 1. Open qBitTorrent on your server.
- 2. Navigate to **Tools** → **Options** (or **Edit** → **Preferences** on Linux).
- 3. Select **Web UI** from the left sidebar.
- 4. Check "**Web User Interface (Remote control)**".
- 5. Configure the following:

Setting	Recommended Value	Des
IP Address	* (all interfaces) or specific IP	Whic
Port	8080	Web
Use HTTPS	Enabled	Requ
Username	admin (change this)	Web
Password	Strong unique password	Web

```
# Test from the same machine
curl -k -X POST "https://localhost:8080/api/v2/auth/login" \
  -H "Referer: https://localhost:8080" \
  -d "username=admin" \
  -d "password=yourpassword"

# Expected: "Ok.SID" response with Set-Cookie header
```

1.2 Browser Requirements

Browser	Minimum Version	Download URL
Google Chrome	88	https://www.google.com/chrome/
Mozilla Firefox	109	https://www.mozilla.org/en-US/firefox/new/
Opera	74	https://www.opera.com/
Yandex Browser	21	https://browser.yandex.ru/
Microsoft Edge	88	https://www.microsoft.com/edge

Check your browser version:

Browser	Method
Chrome	Menu (⋮) → Help → About Google Chrome
Firefox	Menu (≡) → Help → About Firefox

1.3 Network Requirements

Requirement	Description
Same network (recommended)	Browser and server on the same LAN for
HTTPS accessibility	Browser must be able to reach the server
Certificate trust	Self-signed certificates may require brow
Firewall rules	Ports 8443 (Boba) and/or 8080 (qBitTorr
No proxy interference	Corporate proxies may need exceptions f

Testing network connectivity:

```
# Replace with your server IP
SERVER_IP="192.168.1.100"

# Test basic connectivity
ping -c 3 $SERVER_IP

# Test port reachability (Boba)
nc -zv $SERVER_IP 8443

# Test port reachability (qBitTorrent)
nc -zv $SERVER_IP 8080

# Test HTTPS endpoint (accepts self-signed certs)
curl -k -I https://$SERVER_IP:8443/api/v1/health

# Test from the browser's context
```

2.1 Chrome (Chrome Web Store)

The recommended installation method for Chrome users.

Step-by-step:

1. Open Google Chrome.
2. Navigate to the Chrome Web Store page for BobaLink:
 - Search "BobaLink" in the Chrome Web Store, OR
 - Direct link: `https://chrome.google.com/webstore/detail/bobalink`
3. Click the "**Add to Chrome**" button.
4. Review the permission request:

Permission	Why It's Needed
Read and change data on websites	Detect magnet links on websites
Storage	Save configuration and queue
Context menus	Add "Send to Boba" right-click menu
Notifications	Show download status toast

5. Click "**Add extension**" to confirm.
6. The BobaLink icon (tea bubble) appears in the toolbar.

If the icon is hidden:

1. Click the extensions icon (puzzle piece) in the Chrome toolbar.
2. Find BobaLink in the list.

1. Download the release ZIP from GitHub Releases:

```
bash wget
```

```
https://github.com/bobaproject/bobalink/releases/download  
chrome-v1.0.0.zip
```

2. Extract the ZIP to a permanent location:

```
bash unzip bobalink-chrome-v1.0.0.zip -d /opt/bobalink/
```

Important: Do not delete this folder while the extension is installed
directory.

3. Open Chrome and navigate to `chrome://extensions`.
4. Enable "**Developer mode**" using the toggle in the top-right corner.
5. Click "**Load unpacked**".
6. Select the extracted folder (`/opt/bobalink/chrome`).
7. BobaLink appears in the extensions list with its extension ID.
8. (Optional) Note the extension ID for debugging:

```
Extension ID: abcdefghijklmnopqrstuvwxyzabcdef (example)
```

2.3 Firefox (Add-ons Mozilla.org — AMO)

Step-by-step:

1. Open Mozilla Firefox.
2. Navigate to `https://addons.mozilla.org/en-US/firefox/addon`

5. Click "**Add**" to confirm.
6. BobaLink is installed and the icon appears in the toolbar.

2.4 Firefox (Manual Installation)

For environments without AMO access.

Step-by-step:

1. Download the Firefox XPI or ZIP from GitHub Releases:

```
bash wget
```

```
https://github.com/bobaproject/bobalink/releases/download
```

```
firefox-v1.0.0.zip
```

2. Extract to a permanent location:

```
bash unzip bobalink-firefox-v1.0.0.zip -d /opt/bobalink
```

3. Open Firefox and navigate to `about:debugging`.

4. Click "**This Firefox**" on the left sidebar.

5. Click "**Load Temporary Add-on...**".

6. Navigate to the extracted folder and select `manifest.json`.

7. BobaLink loads as a temporary extension.

Note: Temporary extensions are removed when Firefox closes. For permanent installation in Firefox, the extension must be signed by Mozilla or installed via enterprise.

```
// Linux: /etc/firefox/policies/policies.json
// Windows: C:\Program Files\Mozilla Firefox\distribution\policies.json
// macOS: /Applications/Firefox.app/Contents/Resources/distribution/policies.json

{
  "policies": {
    "Extensions": {
      "Install": [
        "https://your-server/bobalink-firefox-v1.0.0-signed.xpi"
      ],
      "Locked": false
    }
  }
}
```

2.5 Opera (Opera Addons)

1. Open Opera Browser.
2. Navigate to <https://addons.opera.com/en/extensions/details/>
3. Click **"Add to Opera"**.
4. Confirm the installation when prompted.
5. The icon appears in the Opera toolbar.

2.6 Opera (Manual Installation)

Opera is Chromium-based and can install Chrome extensions.

1. Enable the "Install Chrome Extensions" feature:
 - Go to <opera://extensions>.

2.7 Yandex Browser (Manual Installation)

Yandex Browser does not have a dedicated extension store for BobaLink. Use

Step-by-step:

1. Download the Chrome release ZIP from GitHub.
2. Extract to a permanent location.
3. Open Yandex Browser and navigate to `browser://extensions`.
4. Enable "**Developer mode**".
5. Click "**Load unpacked**".
6. Select the extracted Chrome build folder.
7. BobaLink installs and the icon appears in the toolbar.

2.8 Chromium (Developer Mode)

For ungoogled-chromium and other Chromium derivatives:

1. Download the Chrome release ZIP.
2. Extract to a permanent location.
3. Navigate to `chrome://extensions`.
4. Enable "**Developer mode**".
5. Click "**Load unpacked**" and select the extracted folder.

Command-line installation (advanced):

```
# Launch Chromium with the extension pre-loaded
```

3.1 Download Release Package

1. Visit the GitHub Releases page:

```
https://github.com/bobaproject/bobalink/releases
```

2. Download the appropriate ZIP for your browser:

Browser	File
Chrome / Opera / Yandex / Edge	bobalink-chrome-v{version}.zip
Firefox	bobalink-firefox-v{version}.zip

3. Verify the download (optional but recommended):

```
```bash
Download the checksum file
wget https://github.com/bobaproject/bobalink/releases/download/v1.0.0/SHA256SUMS

Verify
sha256sum -c SHA256SUMS
Expected: bobalink-chrome-v1.0.0.zip: OK
```
```

3.2 Extract Archive

```
# Create installation directory
sudo mkdir -p /opt/bobalink
```

3.3 Load in Browser

| Browser | Navigation | Action |
|---------|---|---------------------|
| Chrome | <code>chrome://extensions</code> | Enable Developer mo |
| Firefox | <code>about:debugging</code> → This Firefox | Load Temporary Add |
| Opera | <code>opera://extensions</code> | Enable Developer mo |
| Yandex | <code>browser://extensions</code> | Enable Developer mo |
| Edge | <code>edge://extensions</code> | Enable Developer mo |

3.4 Post-Load Verification

After loading:

1. The extension should appear in the extensions list with version 1.0.0.
2. No error messages should appear.
3. The BobaLink icon should be visible in the browser toolbar.

4. Post-Installation Setup

4.1 Open Options Page

4.2 Configure Boba Server URL

For Boba Server mode:

1. In Options → Server Settings, ensure "**Boba Server**" is selected as the
2. Enter your Boba server URL:
 - Docker on same machine: `https://localhost:8443`
 - Docker on LAN server: `https://192.168.1.100:8443`
 - With custom port: `https://your-server:9000`
3. Select "**API Key**" as the authentication method.
4. Enter your API key.
5. Click "**Test Connection**".
6. Wait for the green "Connected" indicator.

For qBitTorrent Direct mode:

1. In Options → Server Settings, select "**qBitTorrent Direct**".
2. Enter your qBitTorrent WebUI URL:
 - Same machine: `https://localhost:8080`
 - LAN server: `https://192.168.1.100:8080`
3. Enter your WebUI username and password.
4. Click "**Test Connection**".
5. Wait for the green "Connected" indicator.

4.3 Run Connection Test

The connection test verifies:

Connection Test

Server: https://192.168.1.100:8443
Mode: Boba Server
Status: Connected
Version: 1.4.2
Latency: 45ms
Auth: OK
API: Compatible

If the test fails:

| Error | Cause | Solution |
|-----------------------|------------------------------|------------------|
| Connection refused | Server not running | Start Boba/qBit |
| Timeout | Firewall or network issue | Check ports, ve |
| Certificate error | Self-signed cert not trusted | Accept certifica |
| Authentication failed | Wrong credentials | Verify API key |
| API incompatible | Server version too old | Upgrade Boba/ |

4.4 Auto-Discover (Optional)

If you don't know your server URL, use auto-discovery:

1. In Options → Server Settings, click "**Discover Servers**".
2. BobaLink will scan your local network for Boba instances.
3. Discovered servers appear in a list with response times

| URL | Timeout | Pu |
|---|---------|----|
| <code>https://boba.local:8443/health</code> | 5s | m |
| <code>https://boba.local:8080/health</code> | 5s | Al |
| <code>https://boba:8443/health</code> | 5s | De |
| <code>https://localhost:8443/health</code> | 3s | Sa |

5. Configuration Guide

5.1 Boba Server Settings

| Setting | Default | Description |
|-----------------------|---------|----------------------------|
| Server URL | (empty) | Full HTTPS URL of your |
| API Key | (empty) | Authentication key from B |
| Connection Timeout | 30s | Maximum wait for API res |
| Health Check Interval | 30s | How often to ping the serv |

Example configuration:

5.2 qBitTorrent Direct Mode (Without Boba)

| Setting | Default | Description |
|-----------------------|---------|-------------------------------|
| Server URL | (empty) | Full HTTPS URL of qBitTorrent |
| Username | (empty) | WebUI username |
| Password | (empty) | WebUI password |
| Connection Timeout | 30s | Maximum wait for API response |
| Health Check Interval | 30s | How often to ping the server |

Note on Referer header: qBitTorrent WebUI requires the `Referer` header to be set. BobaLink handles this automatically.

5.3 Authentication Setup

API Key Authentication (Boba)

1. Obtain an API key from your Boba server administrator.
2. In Options → Server Settings, select "**API Key**".
3. Paste the key into the input field.
4. The key is masked for security. Click the eye icon to reveal.
5. Click "**Test Connection**" to validate.

Basic Authentication (Advanced)

For servers behind reverse proxies with Basic Auth:

1. Select "**Basic Auth**".
2. Enter the Basic Auth username and password.
3. BobaLink will send `Authorization: Basic {base64}` headers.

Custom Header (Advanced)

For reverse proxies requiring custom headers (e.g., Cloudflare Access):

1. Select "**Custom Header**".
2. Enter the header name (e.g., `CF-Access-Client-Id`).
3. Enter the header value.

5.4 Category Configuration

Categories organize torrents in qBitTorrent. To configure:

1. Open Options → Download Preferences.
2. Enter a default category name (e.g., `movies`).
3. Categories are created automatically in qBitTorrent 4.5+.
4. For older versions, pre-create categories in qBitTorrent WebUI.

Common category schemes:

| Category | Use Case |
|----------|----------|
|----------|----------|

5.5 Download Path Settings

Override the default download directory:

- 1. Open Options → Download Preferences.
- 2. Enter an absolute path in **Default Save Path**.

Path examples:

| OS | Example Path |
|--------------------------|-------------------------------|
| Linux | /mnt/downloads/torrents |
| macOS | /Users/username/Downloads/Tor |
| Windows | D:\\Downloads\\Torrents |
| Docker (Linux container) | /downloads/movies |

***Important:** The path must be valid on the **server** (where qBitTorrent runs).*

Using qBitTorrent's Automatic Torrent Management:

- 1. In qBitTorrent WebUI → Options → Downloads.
- 2. Enable "**Automatic Management Mode**".
- 3. Set default save paths per category.
- 4. Torrents are automatically moved to category folders on completion.

2. Navigate to a site with known magnet links. For testing, you can use:
 - `https://ubuntu.com/download/alternative-downloads` (official Ubuntu alternative download page)
 - Any torrent index site
3. Wait for the page to fully load.
4. Look at the BobaLink icon in the toolbar.

Expected behavior:

- The badge shows a number (count of detected torrents).
- The badge is green (connected to server).

6.2 Verify Detection

1. Click the BobaLink icon to open the popup.
2. You should see a list of detected torrents.
3. Each torrent shows:
 - Name (e.g., "Ubuntu 24.04 LTS Desktop")
 - Size (if available)
 - Source (magnet link or .torrent file)

If no torrents are detected:

- Check that the page actually contains magnet links.
- Try scrolling down to trigger lazy loading.
- Check the [Troubleshooting](#) section.

6.3 Verify Sending

1. In the popup, check one or more torrents.
2. Click "**Send Selected**".

Complete verification checklist:

| Check | Method | E |
|---------------------|------------------------------------|---|
| Extension installed | Icon visible in toolbar | Y |
| Server connected | Badge is green | Y |
| Detection works | Badge shows count on torrent page | > |
| Send works | Send a test torrent | S |
| Queue works | Disconnect server, send, reconnect | I |
| Settings persist | Restart browser, check Options | S |

7. Updating

7.1 Automatic Updates from Store

If installed from Chrome Web Store, AMO, or Opera Addons:

- Updates are applied automatically by the browser.
- No user action required.
- Check `chrome://extensions` to see the current version.

7.2 Manual Update Process

chrome://extensions → Find BobaLink → Click refresh icon

Firefox:

about:debugging → This Firefox → Find BobaLink → Click Reload

7.3 Version Compatibility

| BobaLink | Boba Server | qBitTorrent | Status |
|----------|---------------|-------------|--------------------------|
| 1.0.x | 1.0.x – 1.4.x | 4.4+ | Fully compatible |
| 1.0.x | 1.5.x+ | 4.4+ | May not work |
| 1.0.x | < 1.0.x | 4.4+ | Not supported |
| 1.0.x | Any | < 4.4.0 | WebTorrent not supported |

Before updating:

1. Check the release notes for breaking changes.
2. Back up your configuration (export from Options).
3. Ensure your server versions are compatible.

8. Uninstallation

3. Confirm.

8.2 Remove from Firefox

1. Navigate to `about:addons`.
2. Find BobaLink in the Extensions list.
3. Click the `⋮` menu → "**Remove**".
4. Confirm.

8.3 Remove from Opera

1. Navigate to `opera://extensions`.
2. Find BobaLink.
3. Click "**Remove**".
4. Confirm.

8.4 Remove from Yandex

1. Navigate to `browser://extensions`.
2. Find BobaLink.
3. Click "**Remove**".
4. Confirm.

8.5 Clean Up Stored Data

Chrome:

```
// Open DevTools for the extension background page
// Run in Console:
chrome.storage.local.getBytesInUse(null, (bytes) => {
  console.log(`Storage used: ${bytes} bytes`);
});

// Clear all data
chrome.storage.local.clear(() => {
  console.log('All extension data cleared');
});
```

9. Troubleshooting Installation Issues

9.1 "This extension is blocked by the administrator"

Cause: Enterprise policies restrict extension installation.

Solutions:

Option A — Whitelist the extension (IT Admin):

Windows Registry:

HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Google\Chrome\ExtensionInstallForDevs

Add the extension ID

Or via Group Policy:

Administrative Templates → Google Chrome → Extensions →
Configure extension installation allow list

Solution:

1. Ensure you extracted the ZIP completely.
2. Select the folder containing `manifest.json` , not a parent or subfolder.
3. Verify `manifest.json` exists:

```
bash ls /opt/bobalink/chrome/manifest.json
```

9.3 "Invalid value for 'content_security_policy'"

Cause: Browser version too old for MV3 CSP format.

Solution: Upgrade your browser to the minimum required version (Chrome 8

9.4 Extension Loads But Icon Is Gray

Cause: Extension not configured.

Solution:

1. Click the icon to open the popup.
2. Follow the first-time setup wizard.
3. Configure your server URL and credentials.

9.5 "Failed to load extension from..."

Common causes and solutions:

| Error Message | Cause |
|------------------------------------|-----------------------|
| Cannot load extension with file or | macOS metadata folder |

9.6 Certificate Warnings

When using self-signed certificates (common in home lab setups):

Chrome:

1. Navigate to your server URL (e.g., `https://192.168.1.100:8443`).
2. Click "**Advanced**" → "**Proceed to {server} (unsafe)**".
3. The certificate is now trusted for this session.

For permanent trust:

```
# Export the server certificate
openssl s_client -connect 192.168.1.100:8443 </dev/null 2>/dev/null | openssl x509 -outform PEM > boba-cert.pem

# Install system-wide (Linux)
sudo cp boba-cert.pem /usr/local/share/ca-certificates/boba-cert.pem
sudo update-ca-certificates

# macOS
sudo security add-trusted-cert -d -r trustRoot \
  -k /Library/Keychains/System.keychain boba-cert.pem

# Windows (PowerShell admin)
Import-Certificate -FilePath boba-cert.pem \
  -CertStoreLocation Cert:\LocalMachine\Root
```

9.7 Network Timeouts During Setup

| Symptom | Cause | Solution |
|---------|-------|----------|
|---------|-------|----------|

9.8 Firefox-Specific Issues

| Issue | Solution |
|---|---|
| "Add-on could not be installed" (corrupt) | The XPI must be signed. Use <code>about:debugging</code> to install unsigned. |
| <code>chrome.*</code> API errors | The <code>webextension-polyfill</code> may not be loaded. |
| Storage quota exceeded | Firefox has stricter storage limits. Clear old data. |
| Service worker not persistent | Firefox suspends service workers more aggressively. Use <code>ServiceWorkerRegistration.prototype.keepalive</code> to keep alive. |

9.9 Docker-Specific Issues

If running Boba or qBitTorrent in Docker:

```
# Verify container is running
docker ps | grep boba

# Check container logs
docker logs boba-container-name

# Verify port mapping
docker port boba-container-name
# Expected: 0.0.0.0:8443->8443/tcp

# Test from inside the container
docker exec -it boba-container-name curl localhost:8443/health
```

| Issue | Cause | Solution |
|-----------------------------|------------------------------|---|
| Port not exposed | Missing <code>-p</code> flag | Add <code>-p 8443:8443</code> to |
| Container exits immediately | Config error | Check logs with <code>docker</code> |
| Cannot reach from host | Binding to 127.0.0.1 only | Use <code>-p 0.0.0.0:8443</code> |
| SSL errors | Certificate not mounted | Mount cert volume: <code>-v /path/to/certs:/ap</code> |

9.10 Getting Help

If you encounter an issue not covered here:

1. Check the [User Guide Troubleshooting](#) section.
2. Check GitHub Issues: <https://github.com/bobaproject/bobalink>
3. Enable debug logging (see [Developer Guide](#)).
4. Collect the following information for support:
 - Browser and version
 - BobaLink version
 - Server type (Boba/qBitTorrent) and version
 - Error messages (screenshots or text)
 - Steps to reproduce